Storing and Organizing Data

AWS Academy Data Engineering

Introduction

Storing and Organizing Data

Module objectives

This module prepares you to do the following:

- Define storage types that are found in a modern data architecture.
- Distinguish between data storage types.
- Select data storage options that match your storage needs.
- Implement secure storage practices for cloud-based data.

Module overview

Presentation sections

- Storage in the modern data architecture
- Data lake storage
- Data warehouse storage
- Purpose-built databases
- Storage in support of the pipeline
- Securing storage

Lab

- Storing and Analyzing Data by Using Amazon Redshift

Knowledge checks

- Online knowledge check
- Sample exam question

Storage in the modern data architecture

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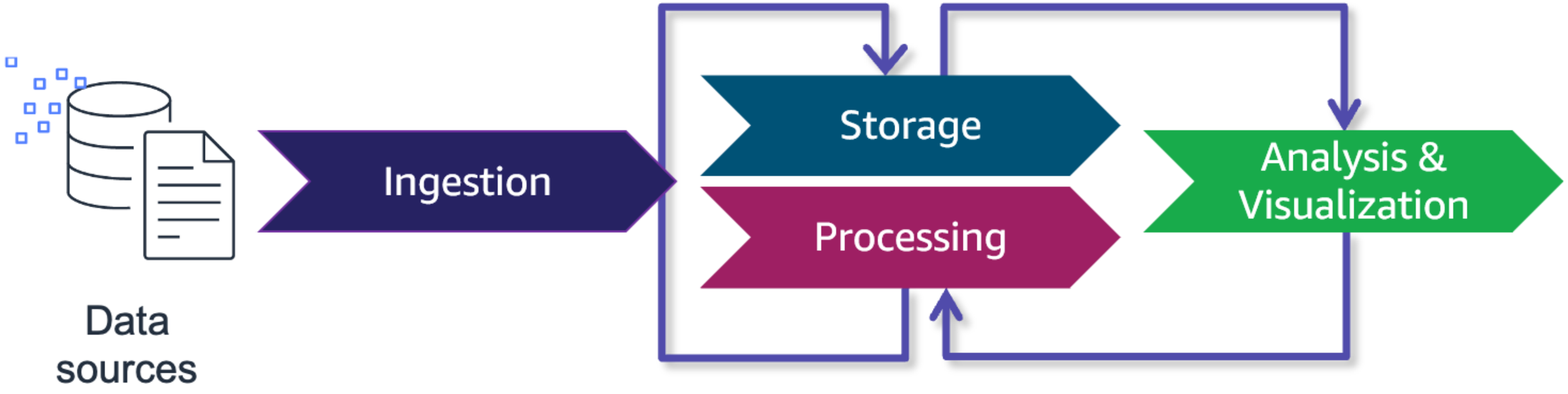
The simplified iterative data pipeline



Storage in the AWS modern data architecture

Types of cloud storage

Block storage

- Offers dedicated, low-latency storage
- Is scalable and offers high performance
- Is similar to local direct attached storage or a storage area network (SAN)
- Example: Amazon Elastic Block Storage (Amazon EBS)

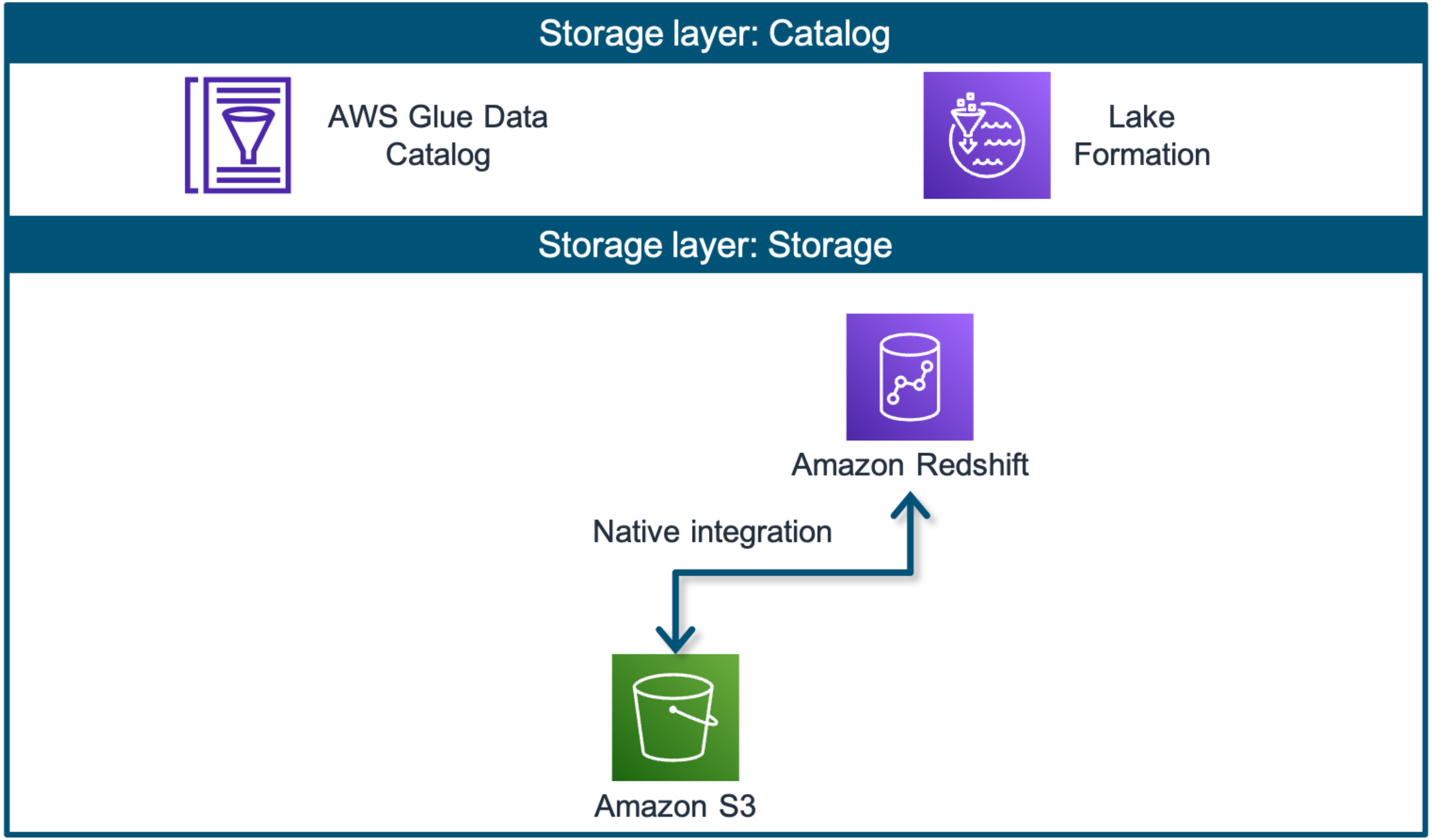
File storage

- Stores data as files
- Is highly scalable
- Is ideal for storage such as content repositories and media stores
- Example: Amazon Elastic File System (Amazon EFS)

Object storage

- Stores unstructured, semistructured, or structured data
- Is highly scalable
- Uses a unique identifier for each object
- Has a lower cost than traditional storage
- Example: Amazon Simple Storage Service (Amazon S3)

Storage layer



Comparing data lakes and data warehouses

Characteristic	Data Warehouse	Data Lake
Data	Relational data from transactional systems, operational databases, and line of business applications	Nonrelational and relational data from Internet of Things (IoT) devices, websites, mobile apps, social media, and corporate applications
Schema	Designed prior to the data warehouse implementation (schema on write)	Written at the time of analysis (schema on read)
Price and Performance	Fastest query results using higher cost storage	Query results getting faster using low-cost storage
Data Quality	Highly curated data that serves as the central version of the truth	Any data, which might or might not be curated (for example, raw data)
Users	Business analysts	Data scientists, data developers, and business analysts (using curated data)
Analytics	Batch reporting, business intelligence (BI), and visualizations	ML, predictive analytics, and data discovery and profiling

Data lake storage

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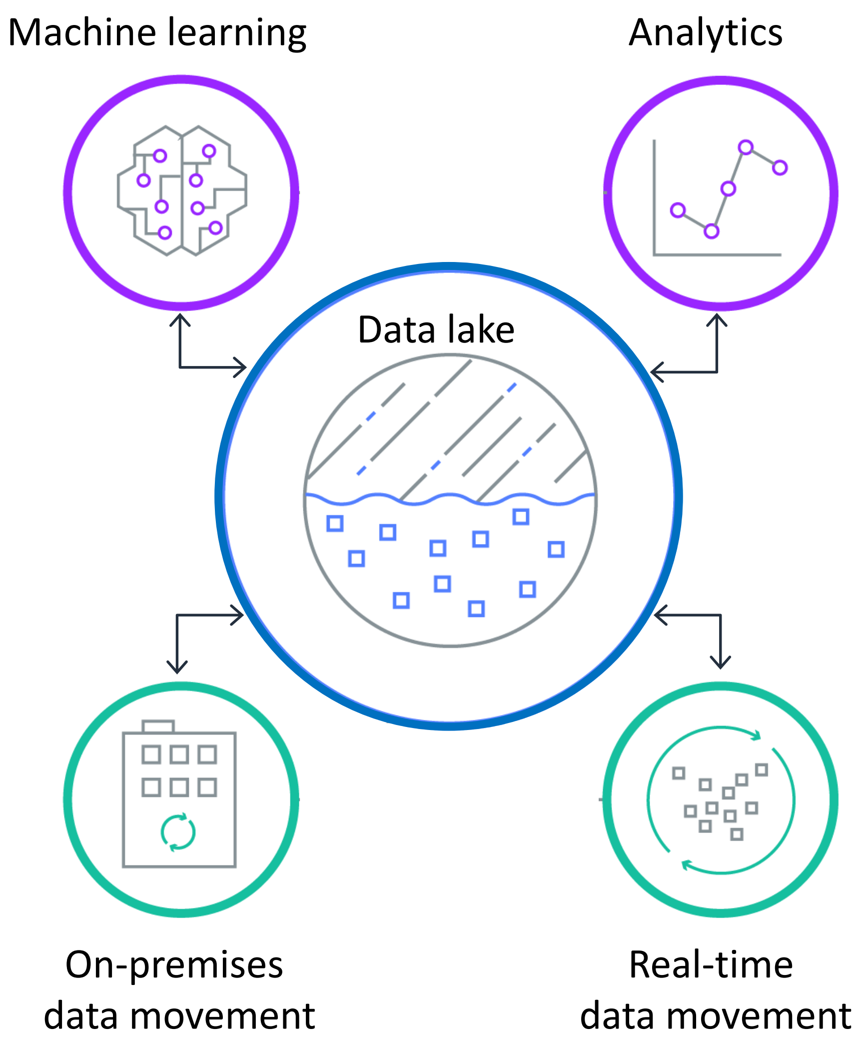
Data lakes

- Provide a centralized repository
- Store both structured and unstructured data
- Catalog and index data for analysis without data movement
- Store, secure, and protect data at unlimited scale
- Offer in-place transformation and querying of data assets
- Are built using Amazon S3



Amazon Simple Storage Service (Amazon S3)

- Is secure, scalable, and durable
- Provides a low-cost storage solution
- Stores structured and unstructured data
- Offers in-place transformation and querying
- Uses object storage classes
- Has a strong data consistency model
- Supports multipart upload
- Is the basis of data lake creation

Amazon S3 storage classes

Amazon S3

General purpose

- S3 Standard

Unknown or changing access

- S3 Intelligent-Tiering

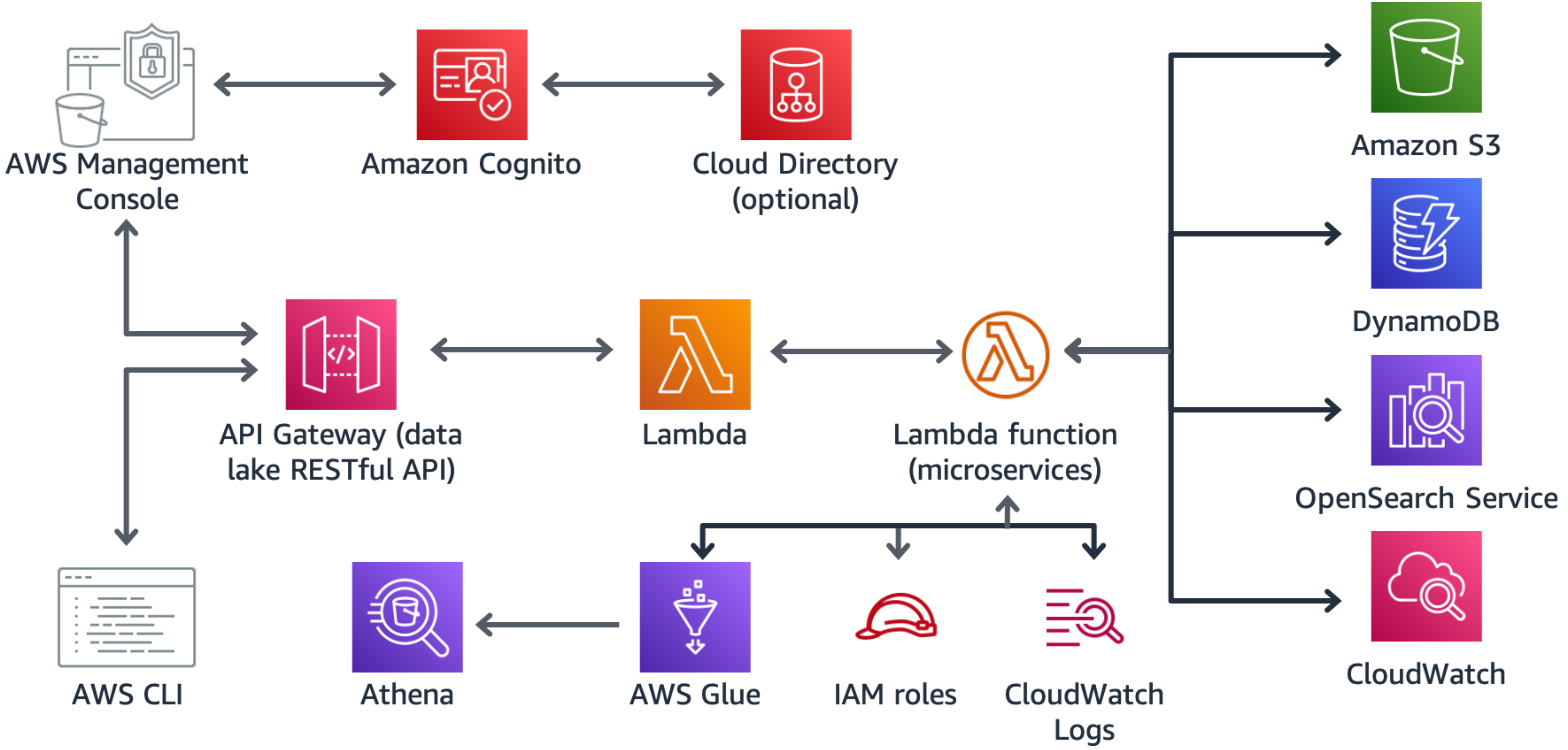
Infrequent access

- S3 Standard-Infrequent Access
- S3 One Zone-Infrequent Access

Archive

- S3 Glacier Instant Retrieval
- S3 Glacier Flexible Retrieval
- S3 Glacier Deep Archive

Example architecture of a data lake



AWS Lake Formation

- Is a fully managed service
- Provides the ability to build, secure, and manage data lakes
- Automates elements of data lake creation
- Augments the AWS Identity and Access Management (IAM) permissions model
- Supports atomic, consistent, isolated, and durable (ACID) transactions by using governed tables
- Integrates with AWS analytics and ML services

Key takeaways: Data lake storage

- Data lakes store data as-is. You don’t need to structure the data before you begin to run analytics.
- Amazon S3 promotes data integrity through strong data consistency and multipart uploads.
- With Lake Formation, you can use governed tables to enable concurrent data inserts and edits across tables.

Data warehouse storage

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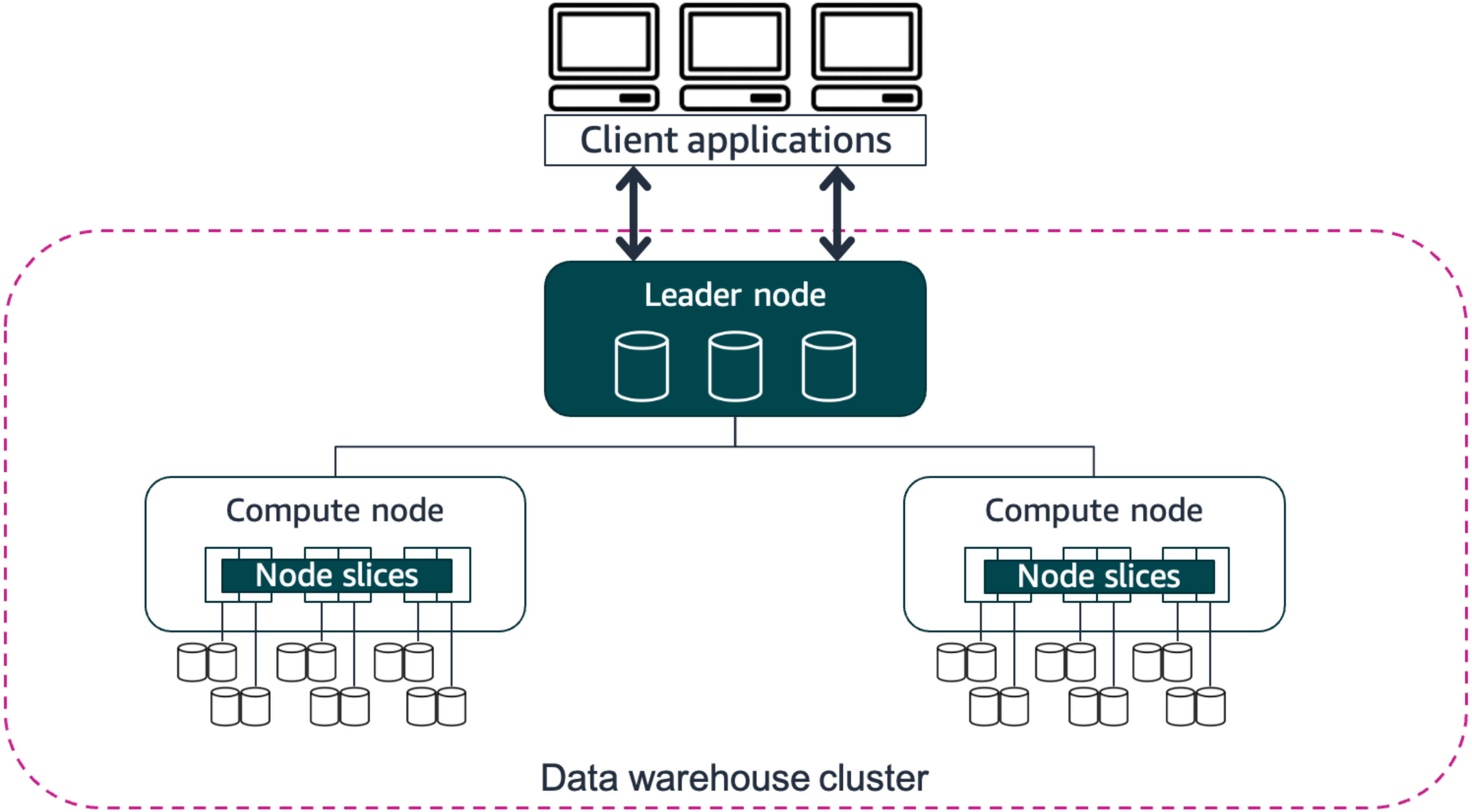
Data warehouses

- Provide a centralized repository
- Store structured and semistructured data
- Store data in one of two ways:
  - Frequently accessed data in fast storage
  - Infrequently accessed data in cheap storage
- Might contain multiple databases that are organized into tables and columns
- Separate analytics processing from transactional databases
- Example: Amazon Redshift

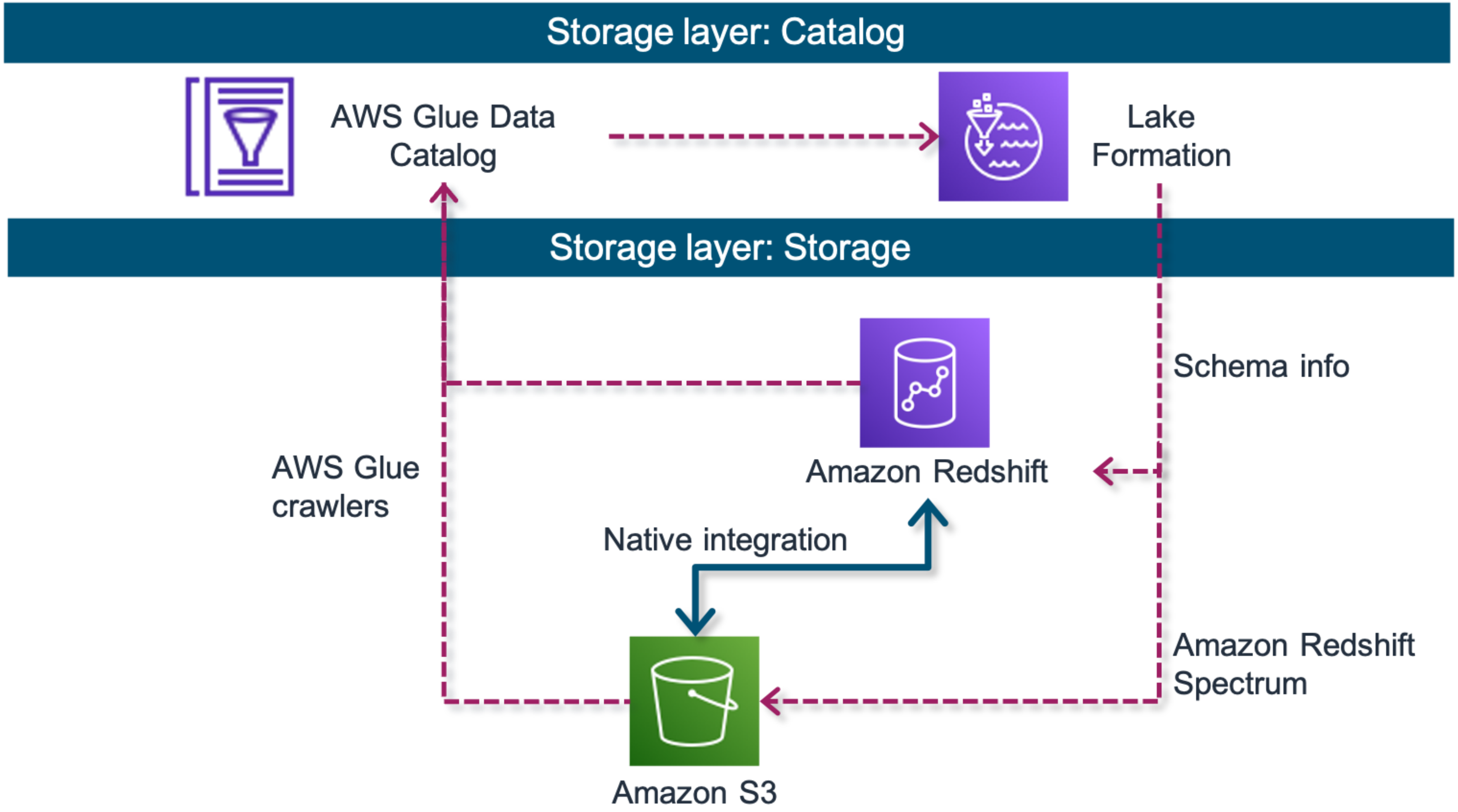
Amazon Redshift

- Provides a cloud-based data warehouse solution
- Is a fully managed service
- Supports near real-time data analysis
- Uses columnar storage
- Offers multiple node types for tailored solutions:
  - DC2
  - DS2
  - RA3

Example architecture: Data warehouse in Amazon Redshift



Amazon Redshift Spectrum



Key takeaways: Data warehouse storage

- Data warehouses consists of three tiers, and can store structured, curated, or transformed data.
- Amazon Redshift is a fully-managed data warehouse services that uses computing resources called *nodes*.
- You can use Redshift Spectrum to write SQL queries that combine data from both your data lake and your data warehouse.

Purpose-built databases

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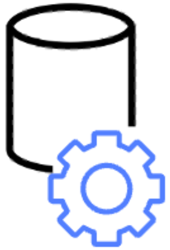
Choosing your purpose-built database

- Choosing the right database is key to supporting your application architecture.
- Your database will affect what your application can handle, how it will perform, and the operation that you are responsible for.
- Consider several factors:
  - Application workload
  - Data shape
  - Performance requirements
  - Operations burden

Factors in choosing a purpose-built database



Application workload

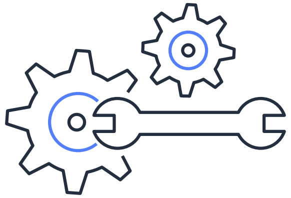
- Is your workload transactional?
- Will your workload be used for analytics purposes?
- Does your workload need caching to improve response times?

Data shape

- How will your data be accessed?
- How often will your data be updated?

Performance

- How fast does your data access need to be?
- What is the average size of the records that you are using?
- How will end users use your service?



Operations burden

- How will you prepare for instance failures?
- How will you configure backups?
- What future upgrades might you need?

Common database use cases

Relational	Key-value	Document	Graph
Traditional applications, enterprise resource planning (ERP), customer relationship management (CRM), ecommerce	High-traffic web applications, ecommerce systems, gaming applications	Content management, catalogs, user profiles	Fraud detection, social networking, recommendation engines
AWS services	**AWS services**	**AWS services**	**AWS services**
Aurora Amazon RDS Amazon Redshift	DynamoDB	Amazon DocumentDB	Neptune

Key takeaways: Purpose-built databases

- Your choice of database will affect what your application can handle, how it will perform, and the operations that you are responsible for.
- When choosing your database, consider several factors:
  - Application workload
  - Data shape
  - Performance
  - Operations burden

Storage in support of the pipeline

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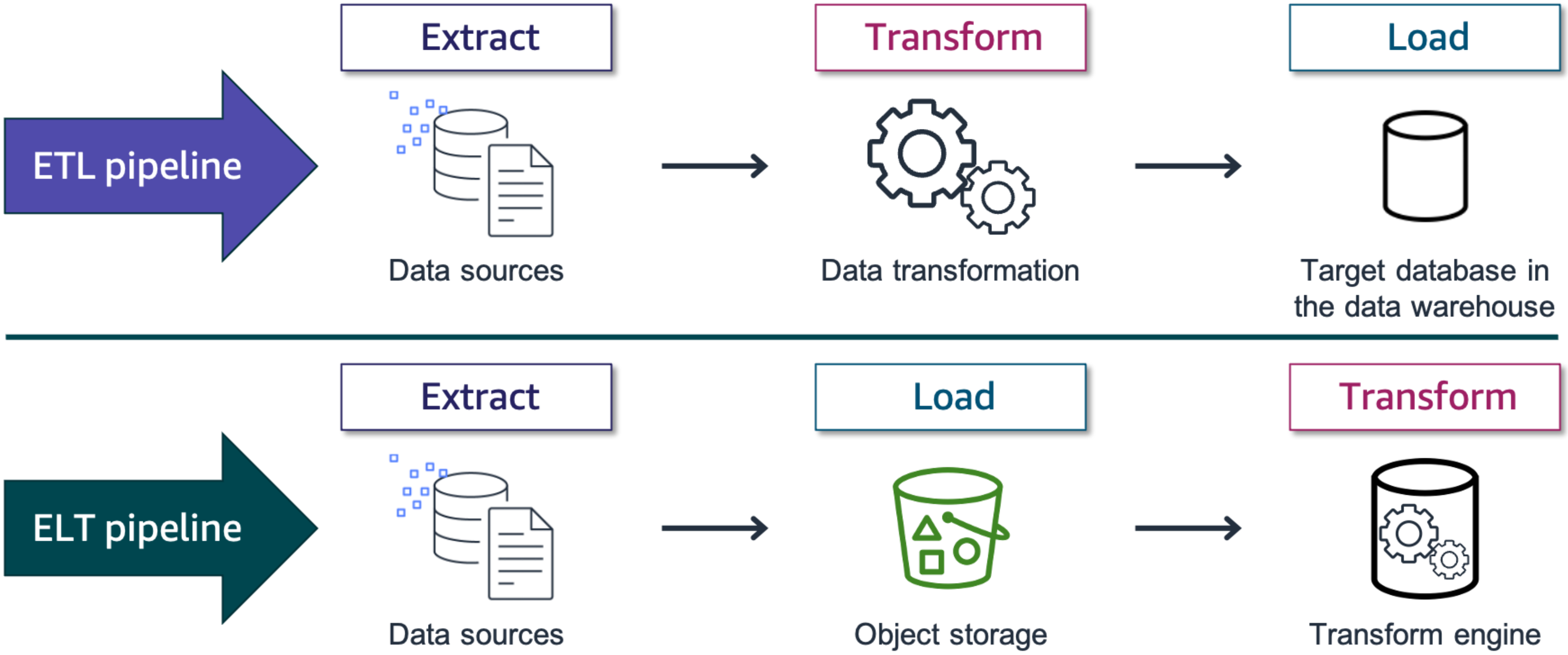
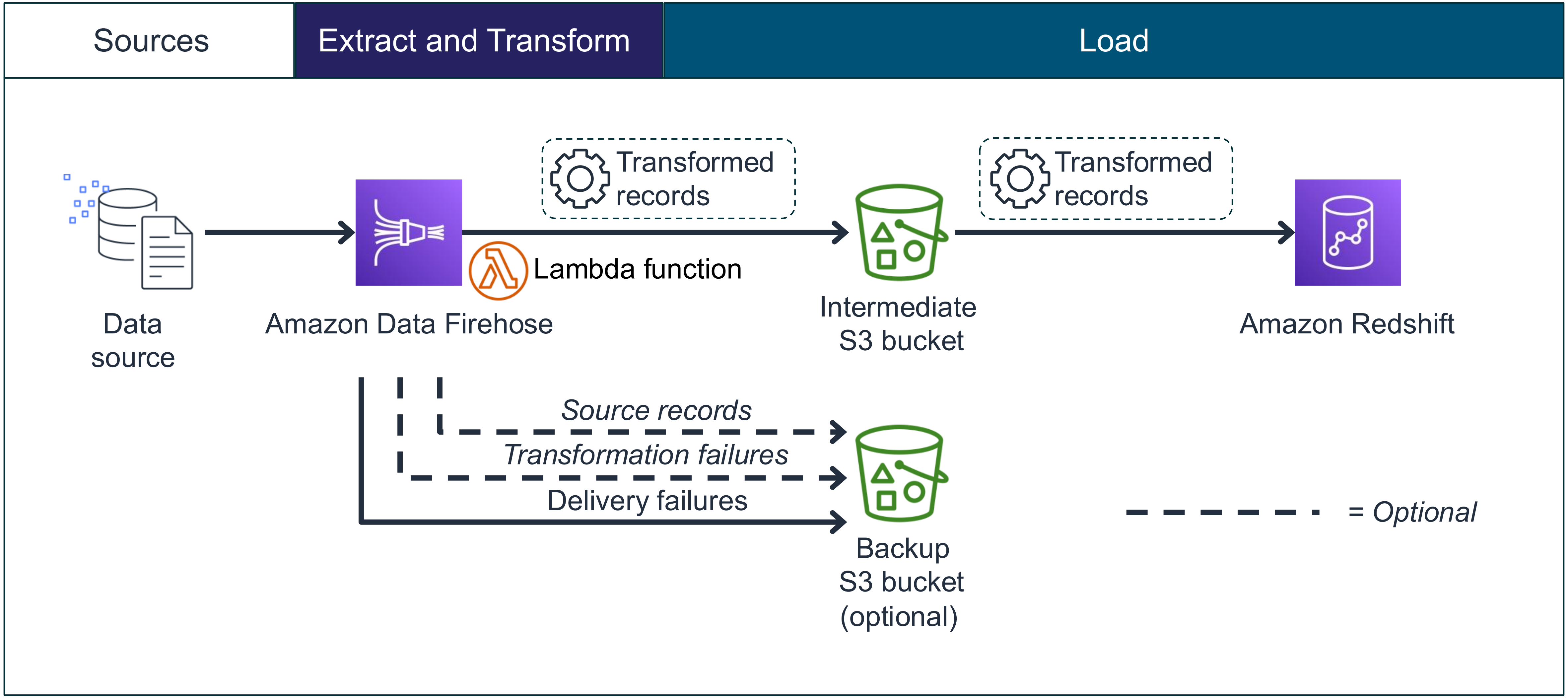
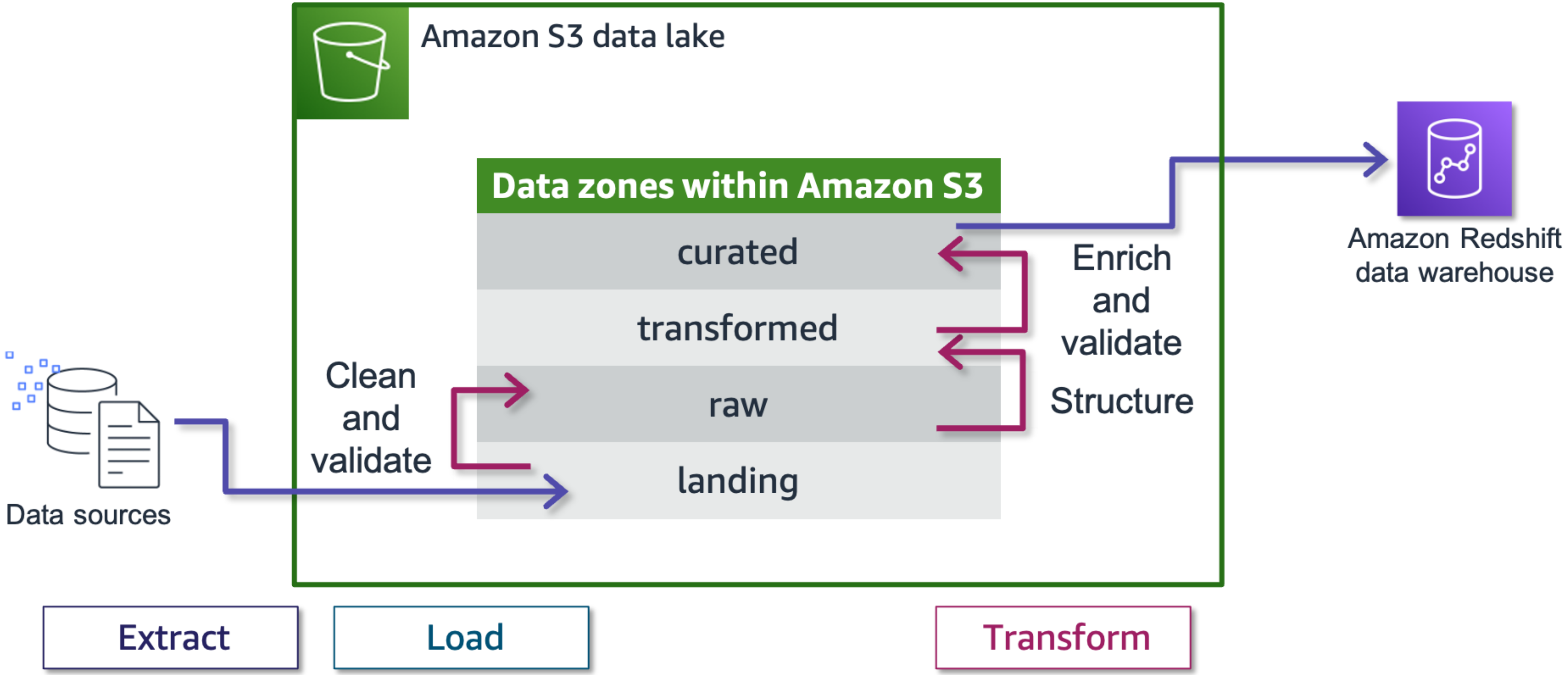
Comparing storage in ETL and ELT pipelines



Example: Storage in support of an ETL pipeline



Example: Storage in support of an ELT pipeline



Key takeaways: Storage in support of the pipeline

- Storage plays an integral part in ELT and ETL pipelines.
- ETL pipelines transform data in buffered memory prior to loading data into a data lake or data warehouse for storage.
- ELT pipelines extract and load data into a data lake or data warehouse for storage without transformation.

Securing storage

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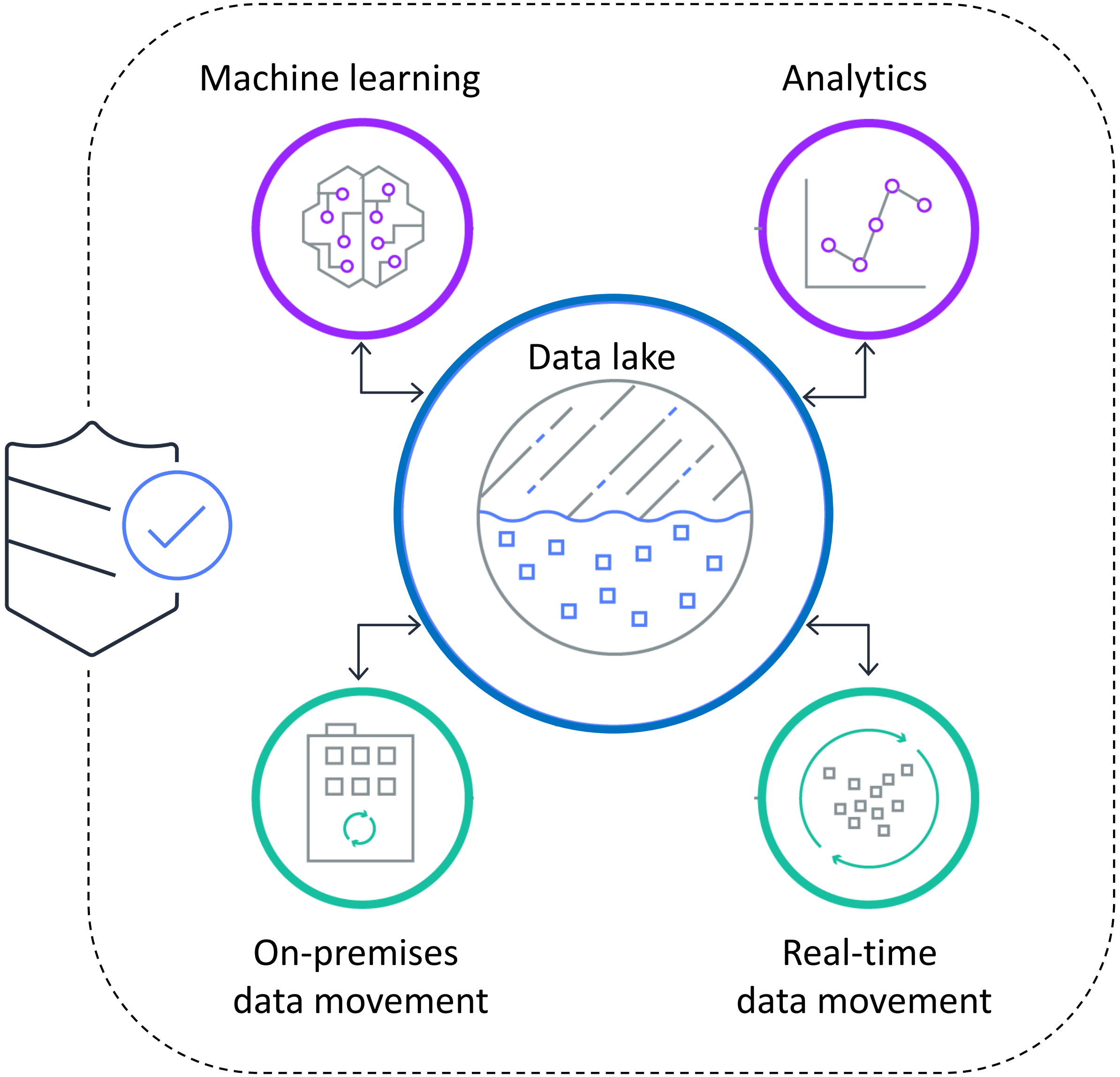
Secure, protect, and manage your AWS data lake

- Benefit from the built-in security of Amazon S3.
- Manage access through resource-based policies and user policies.
- Choose from multiple encryption options.
- Use tags to categorize and manage data and to manage access permissions.
- Use Lake Formation for centralized governance and access control.



Security for a data warehouse in Amazon Redshift

- Amazon Redshift database security is distinct from the security of the service itself.
- Amazon Redshift provides additional features to manage database security.
- Due to third-party auditing, Amazon Redshift can help to support applications that are required to meet international compliance standards.
- Amazon Redshift integrates with Amazon CloudWatch, AWS CloudTrail, and AWS Security Hub for monitoring and alerting.

Key takeaways: Securing storage

- Security for data lake storage is built upon the intrinsic security features of Amazon S3.
- Access policies provide a highly customizable way to provide access to resources in your data lake.
- Data lakes that are built on AWS rely on server-side and client-side encryption.
- Amazon Redshift handles service security and database security as two distinct functions.

Lab: Storing and Analyzing Data by Using Amazon Redshift

Lab introduction: Storing and Analyzing Data by Using Amazon Redshift

- In this lab, you will build a proof of concept to use Amazon Redshift to address the need to query large datasets.
- You will use SQL queries for a music ticket sales dataset. These data files are already stored in Amazon S3 in another account, and you will load them into a Redshift database.
- Open your lab environment to start the lab and find additional details about the tasks that you will perform during this lab.

Debrief: Storing and Analyzing Data by Using Amazon Redshift

- What is a Redshift cluster? What is the purpose of the leader node and the compute nodes in a cluster?
  - Can an AWS admin user change the size or type of a Redshift node?
- Can you use Athena to query a Redshift database?
  - How can you query a Redshift database?
- What skills would a data engineer need to be able to use Amazon Redshift?
- What was the purpose of the MyRedshiftRole IAM role in this lab?

Module wrap-up

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Module summary

This module prepared you to do the following:

- Define storage types that are found in a modern data architecture.
- Distinguish between data storage types.
- Select data storage options that match your storage needs.
- Implement secure storage practices for cloud-based data.

Module knowledge check

- The knowledge check is delivered online within your course.
- The knowledge check includes 10 questions based on material presented on the slides and in the slide notes.
- You can retake the knowledge check as many times as you like.

Sample exam question

A data engineer is designing an infrastructure and wants users to be able to query data directly from files in the company’s data lake, which is built on Amazon S3.

Which service feature would enable this capability?

Identify the key words and phrases before continuing.

The following are the key words and phrases:

- **Query data directly** from files in the company’s **data lake**, which is **built on Amazon S3**
- **Service feature**

Sample exam question: Response options

A data engineer is designing an infrastructure and wants users to be able to **query data directly** from files in the company's **data lake**, which is **built on Amazon S3**.

Which **service feature** would enable this capability?

Choice	Response
A	AWS Lake Formation
B	Amazon Redshift Spectrum
C	AWS Glue crawlers
D	Amazon Neptune graph queries

Sample exam question: Answer

The correct answer is B.

Choice	Response
A	
B	Amazon Redshift Spectrum
C	
D	

Thank you

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